4.1 Rates of change and scales

Definition and results

For $a \neq b$ in f, the rate of change between a and b is $\frac{f(b)-f(a)}{b-a}$. It is the the slope of the slatting through the corresponding points. unit of f divided by that of the variable.

Status and justification. Definition; formula of the slope of a line already studied. $b = a + h$, $h \neq 0$, the rate becomes $\frac{[f(a+h)-f(a)]}{h}$.

Example solved. For $f(x) = x^2$, between 1 and 3, the rate is $\frac{9-1}{3-1}$ = 4. Between 1 and 1 + h, it's worth $2 + h$. The slopes for $h = 0{,}1$ and $h = -0{,}1$ is 2.1 and 1.9.

Point of vigilance

The rate is not $\frac{f(b)}{f(a)}$; h represents a gap, positive or negative depending on the field.

Immediate application. Calculate the rate of $f(x) = x^2 + 3x$ between 2 and 2 + h.

4.2 Derivative number

Definition and results

f is derivable if the rate $\frac{[f(a+h)-f(a)]}{h}$ tends to a real when h tends to 0, in the field. $f\prime(a)$ An intuitive approach to the boundary is used here. kinematics, if f gives a position in meters and time in seconds, the derived number means an instantaneous speed in m s $^{-1}$.

Status and justification. For $f(x) = x^2$, the rate is 2 $a + h$, close to 2 a ; so $f\prime(a) = 2a$. This calculation will be taken as derivative function in Chapter 6.

Example solved. For $s(t) = t^2 + 3t$, the rate around 2 is 7 + h, therefore $s\prime(2) = 7\ \mathrm{m\,s^{-1}}$. The average speed on [2; 3] is 8 m s $^{-1}$; it is not speed at moment 2.

Point of vigilance

We do not replace h by 0 in a non-simplified quotient: we would get a division by 0.

Immediate application. For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

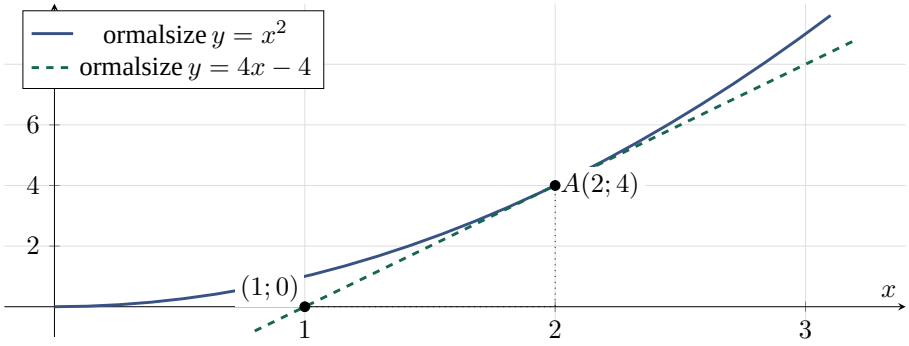
4.3 Tangent and Affine Approximation

Definition and results

If f is derivable from a, the tangent at point ($a; f(a)$ a for equation $y = f(a) + f\prime(a)(x - a)$. For h small, we use $f(a + h) \approx f(a) + f\prime(a)h$. This local approximation does not provide it alone is a bound of error.

Status and justification. The tangent has sloped $f\prime(a)$ and passes through $(a; f(a))$. $y = f\prime(a)x + p$, substitute this point $p = f(a) - af\prime(a)$, hence the equation.

Example solved. For $f(x) = x^2$ in 2, tangent $y = 4 + 4(x - 2) = 4x - 4$. Approach 2 $,01^2$ per 4 + 4(0 $,01) = 4{,}04$; the exact value is 4 $,0401$, deviation $0{,}0001 = h^2$.



The A tangent also passes through (1; 0). Its slope is $\frac{4-0}{2-1}$ = 4. The discontinuous path allows the distinguish from the curve, even in black and white.

Point of vigilance

A tangent can cross the curve elsewhere. "One common point" is not a definition.

Immediate application. We give $f(3)$ = 5 and $f\prime(3) = -2$. Write the tangent and approach $f(3{,}04)$.

4.4 When Differentiability Fails

Definition and results

The square root function is defined on [0; + ∞[but is not derivable in 0. The function absolute value is defined on $\mathbb{R}$ but is not driftable in 0. Be defined or have a curve therefore does not guarantee differentiability.

Status and justification. For $h > 0$, $\frac{\sqrt{h} - 0}{h} = \frac{1}{\sqrt{h}}$, which does not tend to a real finite when h is close to 0. $|x|$, rate $\frac{|h|}{h}$ is 1 to the right and -1 left: no common value.

Example solved. For $|x|$, the slattings from the origin all have slope 1 to the right and -1 à left. The angular dot does not have a single finished slope tangent.
limit direction is vertical; there is no actual derivative number.

Point of vigilance

A very large slope is not proof of non-differentiability; it is necessary to explain the behaviour rate.

Immediate application. Justify that $g(x) = |x - 2|$ is not detrievable in 2.
Differentiability of $x \mapsto \overline{x}$ in 0 by its rate.